

SALES DATA ANALYSIS REPORT

#Problem Statement:

In today's competitive retail environment, businesses need to understand their sales performance to make better decisions. The objective of this project is to analyze a retail sales dataset to identify trends. The main problem is to extract meaningful insights from the sales dataset to improve business growth.

This project aims to solve the following problems:

- 1) Identify which products contribute the most to total revenue.
- 2) Analyze monthly sales trends to understand performance over time.
- 3) Determine the best and worst performing months.
- 4) Detect low-performing products that may need improvement or removal.
- 5) Understand patterns in customer purchasing behavior.

#Methodology:

Data Collection:

The dataset was taken in CSV format containing product names, sales dates, quantity, profit, Borrow and revenue.

Data Wrangling:

- 1) Converted date column into proper format.
- 2) Extracted month from the date.
- 3) Cleaned missing or incorrect values.
- 4) Organized data into categories.

Data Analysis:

- 1) Grouped data by product to calculate total revenue.
- 2) Analyzed monthly revenue trends.
- 3) Identified top-selling products.

Tools Used:

- 1) Python (Pandas , Matplotlib)
- 2) Jupyter Notebook
- 3) Seaborn – for better data visualization
- 4) Microsoft Excel – for initial data viewing

#Visualizations

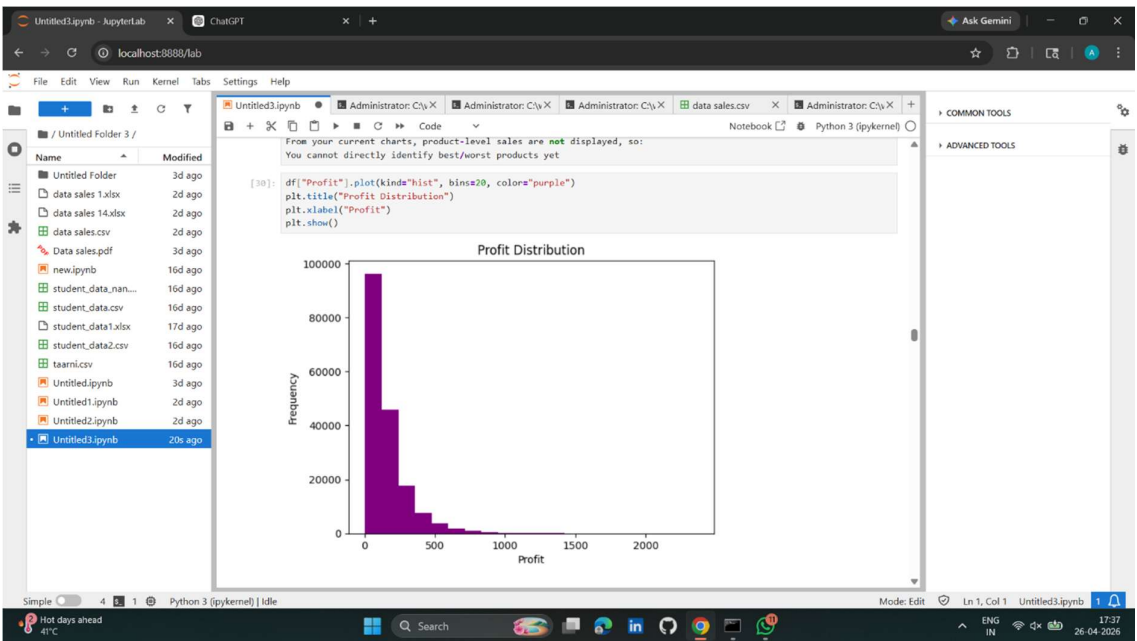
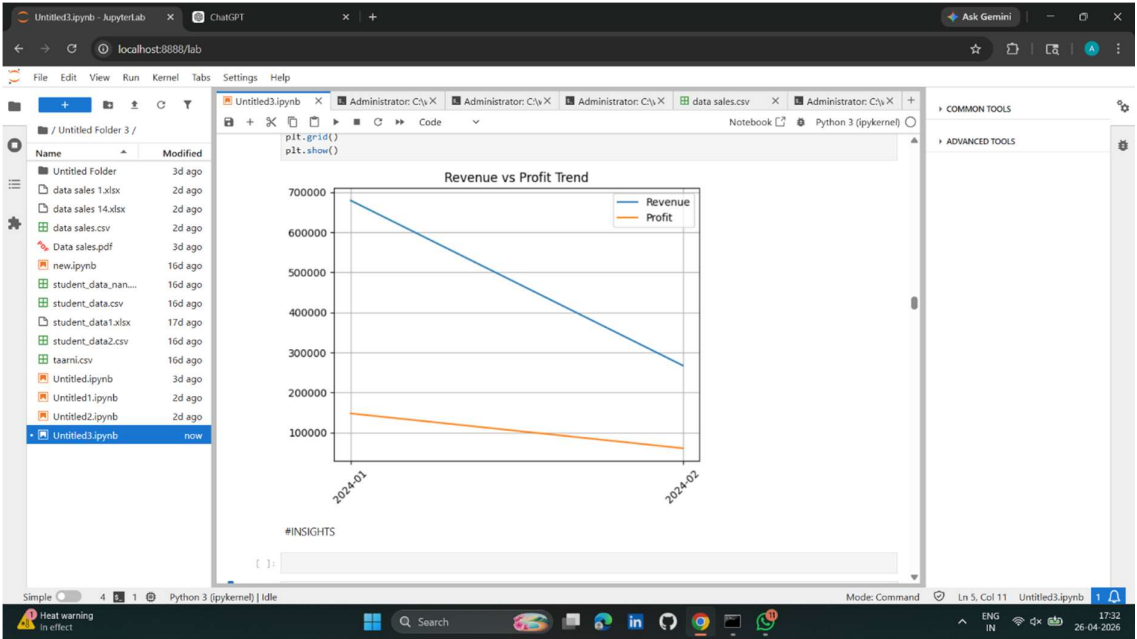
To better understand the data, the following visualizations were created:

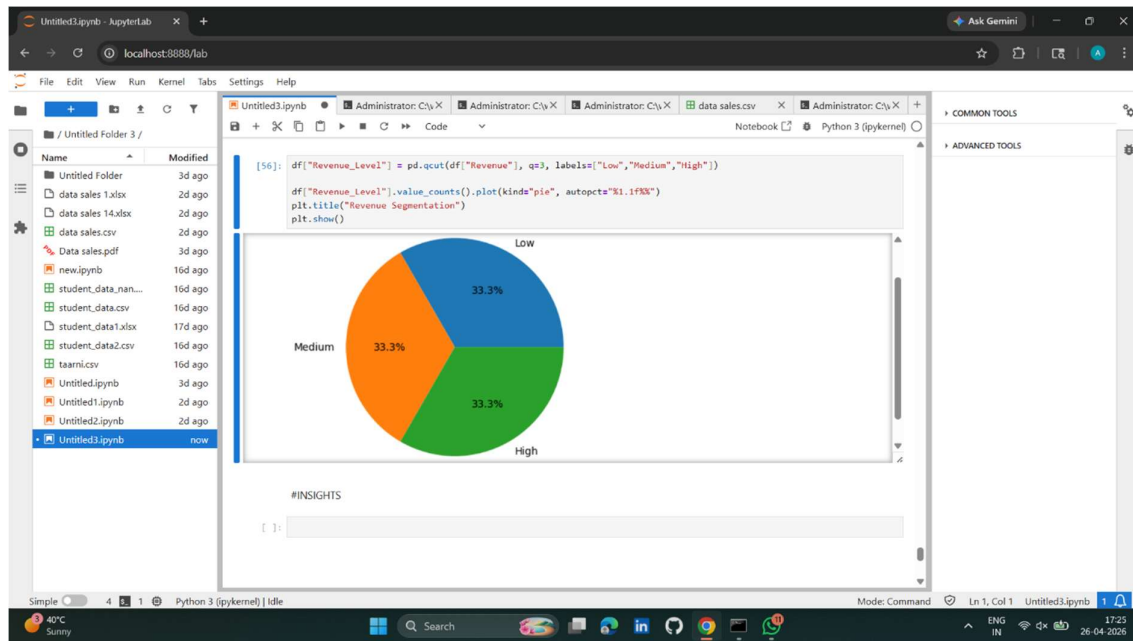
Bar Chart (Top Products)

This chart shows the top-selling products based on revenue. It helps identify which products contribute the most to sales.

Line Chart (Sales Over Time)

This chart represents monthly revenue trends. It shows how sales change over time and highlights patterns.





#Insights:

- 1) **Best Month:** The highest sales were recorded in [Write Month]
- 2) **Worst Month:** The lowest sales were recorded in [Write Month]
- 3) **Top Product:** The product generating maximum revenue was [Product Name]
- 4) **Product Trends:** Some products consistently performed well, while others showed low performance